

HEI LAM (JARVIS)

Software Engineer · AI/ML · Distributed Systems · Full-Stack · iOS

Software Engineering M.S. student at San Jose State University with a Computer Engineering foundation and hands-on experience building distributed AI infrastructure, machine-learning network systems, privacy-focused iOS applications, cybersecurity tools, and cloud-backed web products. Comfortable working across application, systems, networking, and AI layers—from model serving and REST services to native mobile UX and production-oriented frontend architecture.

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PROFESSIONAL PROFILE

Engineering-focused graduate student with experience translating academic concepts into working software systems. Project work spans distributed large-language-model inference, machine-learning-assisted load balancing, phishing and vulnerability detection, iOS privacy/security applications, AI-assisted travel planning, expense-management software, and cloud media platforms. Strongest areas include Python backend development, Swift/iOS development, React-based frontend engineering, applied machine learning, computer networking, cybersecurity, Docker-based environments, and cloud deployment. Known for taking ownership of end-to-end implementation, debugging across layers, and explaining technical systems clearly to teammates and users.

EDUCATION

Master of Science in Software Engineering	Jan 2024 — Present
San Jose State University · Charles W. Davidson College of Engineering · San Jose, CA	
Bachelor of Science in Computer Engineering	Aug 2020 — Dec 2024
San Jose State University · Charles W. Davidson College of Engineering · San Jose, CA	

ALGORITHMS & SYSTEMS Advanced Algorithms, Data Structures, Operating Systems, Computer Architecture, Compiler Design, Real-Time Embedded Systems	NETWORKS & SECURITY Computer Networks, Network Security, Information Security, Software Security Technologies
SOFTWARE & DATA Software Systems Engineering, Enterprise Software Platforms, Database Systems	APPLICATION ENGINEERING Mobile Software Engineering and project-based full-stack, cloud, AI, and distributed-systems development

TECHNICAL EXPERTISE

Languages	Python, Java, C, C++, JavaScript, Swift, SQL, HTML, PHP, Verilog
Web & APIs	React, Vite, Node.js, Express.js, FastAPI, REST APIs, React Router, Tailwind CSS
AI / ML	scikit-learn, Hugging Face, LM Studio, local LLM integration, Isolation Forest, regression models, anomaly detection, model-serving workflows
iOS / Apple	SwiftUI, SwiftData, MapKit, Vision, VisionKit, Swift Charts, LocalAuthentication, PDFKit, PhotosUI, UIKit, Xcode
Cloud & DevOps	AWS, Docker, Docker Compose, Terraform, Firebase Authentication, Firestore, Cloudinary, Vercel, nginx
Systems & Networks	Linux/Ubuntu, macOS, Windows, distributed systems, OpenFlow, Mininet, Ryu, network security, computer architecture
Data & Storage	MySQL, MongoDB, Firestore, pandas, CSV-based data pipelines, safetensors model weights
Developer Tools	Git, GitHub, VS Code, Android Studio, pytest, shell scripting, LM Studio, command-line debugging

SELECTED TECHNICAL PROJECTS

Lumina — Distributed Split LLM Inference

SJSU · 2026

Python · Hugging Face · PyTorch · FastAPI/REST · Docker · AWS · Terraform · React/Vite · nginx

GitHub ↗

- Built a distributed inference architecture that partitions transformer layers across **three physically separate machines**, with head, middle, and tail nodes exchanging serialized hidden states over REST services.
- Implemented a tracker service for node registration, heartbeats, liveness monitoring, request tracing, assignment versioning, and dynamic layer-boundary calculation based on each node's available VRAM.
- Implemented memory-aware CPU model loading using PyTorch's meta device and direct safetensors access so each node materializes only the embedding/layers/output components it executes instead of loading an entire model.
- Supported multiple transformer families, including Qwen2/Qwen2-MoE, GPT-2, and Phi-3/Phi-4 Mini, while keeping model-specific behavior behind an adapter layer.
- Containerized the multi-service topology with Docker Compose; created AWS-oriented deployment assets with EC2/Terraform configurations and an nginx reverse proxy.
- Developed a React/Vite operations dashboard for text generation, node health, CPU/RAM/VRAM visibility, cluster topology, request traces, and client-side logs.

ML-Based Software-Defined Network Load Balancer

SJSU · Spring 2026

Python · Ryu · Mininet · OpenFlow 1.3 · scikit-learn · pandas · Linux · iperf3

GitHub ↗

- Designed an SDN test environment in Mininet using an OpenFlow switch, a Ryu controller, one client, and three backend servers to evaluate programmable load-balancing strategies.
- Implemented a virtual-IP Round Robin controller with DNAT/SNAT and fast-path flow installation, then developed a Least-Loaded policy based on active-flow counts.
- Built a data-collection and training pipeline that evaluates Decision Tree, Random Forest, and Ridge Regression models using RMSE and saves the best model for runtime use.
- Integrated the trained model into an adaptive Ryu controller that predicts backend latency from traffic rate, active flows, and load imbalance, selecting the server with the lowest predicted latency while applying balancing penalties.
- Created repeatable latency, throughput, ping, and packet-drop benchmark scripts and summarized results for comparison across routing strategies.

PixelVault — AI-Enhanced Cloud Media Platform

SJSU · 2026

React · Vite · Tailwind CSS · Firebase Auth · Firestore · Cloudinary · Hugging Face · Vercel

[GitHub ↗](#)

- Co-developed a responsive single-page media-management platform that combines cloud photo/video organization with AI-assisted editing workflows.
- Integrated Firebase Authentication and protected routes for account workflows, Firestore for cloud-backed application data, and Cloudinary for scalable media upload and delivery.
- Built gallery search/filtering, custom albums, multi-select workflows, drag-and-drop ordering, comments/likes, profile management, and responsive dark/light interfaces with reusable React components.
- Integrated Hugging Face inference into a preview-before-apply, non-destructive AI photo-retouching flow and deployed the application on Vercel.

ScamShield — Privacy-Focused iOS Scam Detection

Personal Project · 2026

Swift · SwiftUI · SwiftData · Vision/VisionKit · LocalAuthentication · PDFKit · Combine

[GitHub ↗](#)

- Developed a native iOS security application that analyzes suspicious text, URLs, phone numbers, screenshots, and camera-scanned documents for phishing and scam indicators.
- Designed an explainable rule-based risk engine that breaks scores down across suspicious URLs, credential/payment requests, brand impersonation, threat language, urgency, secrecy, remote-access requests, and recognized scam patterns.
- Integrated Vision OCR and VisionKit document scanning to extract text from screenshots and physical documents before local risk analysis.
- Implemented local scan-history persistence and search/filtering with SwiftData, optional Face ID/Touch ID/passcode protection, and PDF security-report generation for sharing or printing findings.
- Designed the product around privacy-first principles: no required account, no advertising/tracking, local OCR, and on-device analysis whenever possible.

SplitMate — Native iOS Expense-Splitting Application

Personal Project · 2026

Swift 6 · SwiftUI · SwiftData · Swift Charts · Vision · PDFKit · PhotosUI

[GitHub ↗](#)

- Built a native expense-sharing application for friends, roommates, classmates, and travel groups with persistent group, member, expense, payment, and settings models.
- Implemented balance calculations that compute each participant's share, maintain member net balances, and simplify debts into clear "who owes whom" settlement recommendations.
- Added restaurant-specific splitting with subtotal, tax, tip, automatic total calculation, equal splitting, and per-person costs.
- Integrated Vision OCR for attached receipt images, Swift Charts for spending/category analytics, configurable appearance/currency settings, and PDF group-summary generation.
- Structured the application with reusable views/components and dedicated services for balance calculations, statistics, restaurant splitting, OCR, PDF rendering, and user settings.

Lemon Travel Planner — AI-Assisted iOS Itinerary Planner

Personal Project · 2026

Swift · SwiftUI · SwiftData · MapKit · LM Studio · REST APIs · Local LLMs

[GitHub ↗](#)

- Developed an iOS travel-planning application for trip creation, day-by-day itinerary generation, activity editing, transportation records, cost estimation, and persistent trip storage.
- Integrated a locally hosted language model through LM Studio's OpenAI-compatible chat-completions endpoint, enabling AI itinerary generation without requiring a cloud-only workflow.
- Built provider-switching configuration for mock, local LM Studio, and remote/OpenAI-style backends to separate product features from the underlying AI service.

- Integrated MapKit-based place search and route calculation so generated plans can be connected to real locations and map-based trip views.
- Organized the codebase around distinct models, services, and feature modules to keep itinerary generation, place lookup, routing, and UI responsibilities maintainable.

Phishing Detection System

SJSU · Fall 2025

Python · scikit-learn · Isolation Forest · LM Studio · Tkinter · Data Analysis

- Built a phishing-detection application that processed approximately **517,000 Enron emails** to identify suspicious messages using a hybrid detection approach.
- Combined heuristic/rule-based scoring with Isolation Forest anomaly detection to surface unusual or high-risk email patterns.
- Integrated a locally hosted LLM through LM Studio to supplement model/rule outputs with more human-readable phishing-risk explanations.
- Developed a Tkinter desktop interface for CSV ingestion, scan execution, risk display, and explanation review.

AI-Assisted C/C++ Vulnerability Analyzer

SJSU · Spring 2025

C · C++ · Python · Local LLMs · Static Analysis · Software Security

- Developed an AI-assisted source-code analysis workflow for identifying memory-safety and concurrency risks in C/C++ programs.
- Focused analysis on security-relevant patterns such as buffer overflows, unsafe memory operations, and race-condition risks.
- Used locally hosted LLM prompts to produce diagnostic explanations and prioritized remediation guidance rather than returning only raw warnings.
- Evaluated the workflow against multiple open-source code samples to identify unsafe patterns and support manual security review.

Restaurant Booking Web Application

SJSU · Spring 2025

JavaScript · React · Node.js · Express.js · Vite · REST APIs

- Developed and deployed an OpenTable-style full-stack reservation application with distinct workflows for customers, business owners, and administrators.
- Implemented REST APIs with Node.js/Express and React interfaces for restaurant discovery, reservations, account flows, and restaurant-management tasks.
- Refactored frontend component/build organization and optimized Vite delivery, improving page-load performance by approximately **30%** while making feature maintenance easier.

LEADERSHIP & CAMPUS ENGAGEMENT

Activities Coordinator — Chinese Campus Fellowship

Aug 2025 — Present

San Jose State University

- Coordinate cultural and community events with student organizations, including cross-organization Mid-Autumn Festival programming with the Hong Kong Student Association.
- Manage event planning details spanning promotion, logistics, activity design, participant coordination, and execution.
- Plan and host Praise & Worship Night, coordinating program flow, rehearsal schedules, communication, and on-stage execution.
- Organize recurring social activities such as billiards and bowling nights to strengthen participation and member engagement.

ENGINEERING STRENGTHS

Systems Thinking
Comfortable tracing behavior across networking, APIs, model execution, storage, and UI layers.

End-to-End Ownership
Builds projects from architecture and implementation through testing, deployment, documentation, and demos.

Applied AI
Uses ML/LLMs as engineering components inside measurable, user-facing workflows rather than standalone demos.

LANGUAGES & ADDITIONAL TOOLS

Spoken Languages	Cantonese (Native) · Mandarin (Native) · English (Fluent)
Productivity	Microsoft Office · Google Workspace · Apple iWork
Creative	Final Cut Pro